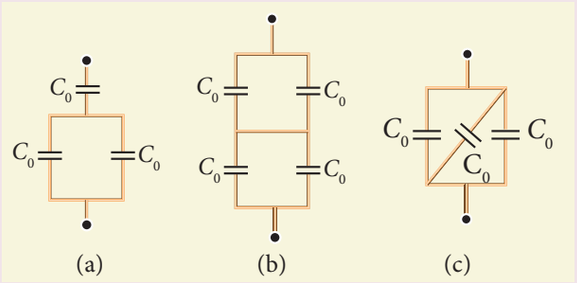
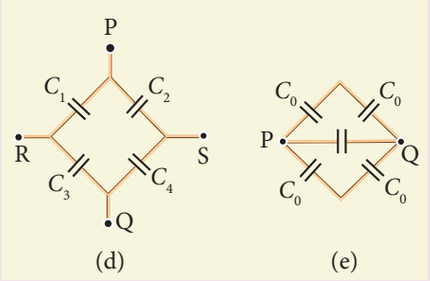
Ans: $\Delta U = +1.12$ J, positive sign implies that to move the charge $-2\mu C$ external work is required.

11. Calculate the resultant capacitances for each of the following combinations of capacitors.





Ans: (a) $\frac{2}{3}C_o$ (b) C_o (c) $3C_o$

(d) across PQ:

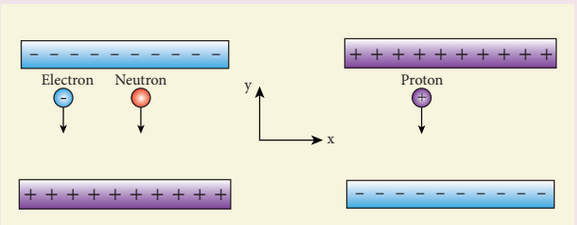
$$\frac{C_1C_2C_3 + C_2C_3C_4 + C_1C_2C_4 + C_1C_3C_4}{(C_1 + C_3)(C_2 + C_4)}$$

across RS:

$$\frac{C_1C_2C_3 + C_2C_3C_4 + C_1C_2C_4 + C_1C_3C_4}{(C_1 + C_2)(C_3 + C_4)}$$

(e) across PQ: $2\,C_o$

12. An electron and a proton are allowed to fall through the separation between the plates of a parallel plate capacitor of voltage 5 V and separation distance h = 1 mm as shown in the figure.



(a) Calculate the time of flight for both electron and proton (b) Suppose if a neutron is allowed to fall, what is the time of flight? (c) Among the three, which one will reach the bottom first? (Take $m_p = 1.6 \times 10^{-27}$ kg, $m_e = 9.1 \times 10^{-31}$ kg and g = 10 m s^{-2})

Ans:

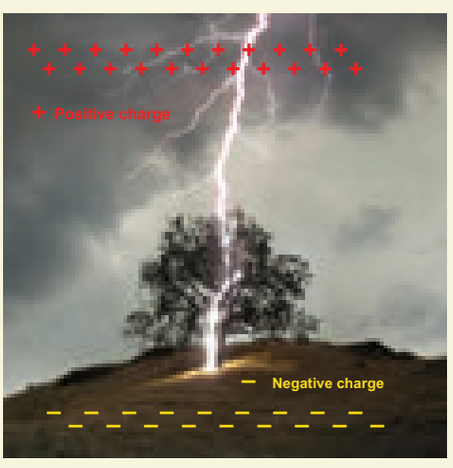
(a) $t_e = \sqrt{\frac{2hm_e}{eE}} \approx 1.5$ ns (ignoring the gravity),

$t_p = \sqrt{\frac{2hm_p}{eE}} \approx 63$ ns (ignoring the gravity)

(b) $t_n = \sqrt{\frac{2h}{g}} \approx 14.1$ ms

(c) electron will reach first

13. During a thunder storm, the movement of water molecules within the clouds creates friction, partially causing the bottom part of the clouds to become negatively charged. This implies that the bottom of the cloud and the ground act as a parallel plate capacitor. If the electric field between the cloud and ground exceeds the dielectric breakdown of the air (3×10^6 Vm^{-1}), lightning will occur.



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